

Requirements and Specifications

Project: RoomRadar QC

Purpose

This document defines the main requirements for RoomRadar QC and connects them to the user discovery interviews. The goal is to make the requirements clear, testable, and useful for planning QA tests.

1. Project Overview

RoomRadar QC is a web application that helps Queens College students find rooms that may be free based on class schedule data.

Users can search by:

- Day
- Start time
- End time
- Building

The system returns rooms that do not have a scheduled class conflict during the selected time. RoomRadar QC does not guarantee live room availability. A room may still be locked, occupied, reserved, restricted, or unavailable even if no class is scheduled.

2. User Discovery Findings

The team interviewed Queens College students to understand how they find study spaces on campus. The interviews showed that students often walk around campus checking rooms

manually, which can waste time and cause frustration.

Finding ID	Interview Finding
DF-1	Students often struggle to find quiet or empty spaces on campus.
DF-2	Students frequently walk around buildings or check rooms manually. Some students said this can take 30 to 40 minutes.
DF-3	Many students said they would use a website or app that shows rooms that may be open.
DF-4	Students want to search for rooms available now and also plan for future times.
DF-5	Building/location filtering is useful because students want rooms near their classes or current location.
DF-6	Students want to know how long a room may remain usable before another class begins.
DF-7	Students want the system to be quick and convenient, especially on mobile.
DF-8	Students were generally not aware that raw course schedule data exists online or in CSV format.
DF-9	Some students suggested future features such as live occupancy, check-ins, reservations, room pictures, notifications, and room type information.
DF-10	Schedule-based availability is useful, but students also mentioned locked rooms, occupied rooms, and spaces unavailable for non-class reasons.

3. Stakeholders

Stakeholder	Interest
Students	Want a faster way to find possible rooms for studying, relaxing, or working.
Admin user	Needs to upload semester schedule data.
Development team	Needs a maintainable frontend, backend, database, and deployment setup.
QA/reviewer	Needs requirements that can be tested and traced.
Instructor/evaluator	Needs evidence of software engineering process, architecture, testing, and documentation.

4. Stakeholder Requirements

ID	Stakeholder Requirement	Based On
SR-1	Students need a faster way to find rooms that may be usable between classes.	DF-1, DF-2, DF-3
SR-2	Students need to search by day and time.	DF-4
SR-3	Students need to filter by building/location.	DF-5
SR-4	Students need to know how long a room may remain usable.	DF-6
SR-5	Students need a quick and simple interface that works on phone or laptop.	DF-7
SR-6	Students need a dedicated interface because raw schedule data is not easy to find or use.	DF-8
SR-7	Admin users need a controlled way to upload semester schedule data.	Project need
SR-8	Users need to understand that results are schedule-based suggestions, not guaranteed live availability.	DF-10
SR-9	Admin features must be protected from unauthenticated users.	Security need
SR-10	Future versions should consider live occupancy, check-ins, reservations, room pictures, notifications, and room type details.	DF-9

5. Functional Requirements

ID	Functional Requirement	Related SR	Verification
FR-1	The system shall provide a room search page with day, start time, end time, and building inputs.	SR-1, SR-2, SR-3	UI test
FR-2	The system shall require users to select day, start time, and end time before searching.	SR-5	Validation test
FR-3	The system shall reject searches where start time is greater than or equal to end time.	SR-5	Validation test
FR-4	The frontend shall send room search requests to the backend API.	SR-1, SR-2	API integration test
FR-5	The backend shall return rooms without scheduled	SR-1, SR-2	API test

	class conflicts for the selected day and time.		
FR-6	The system shall allow users to filter search results by building.	SR-3	UI/API test
FR-7	The system shall display results in readable room cards.	SR-5	UI test
FR-8	Each room result shall show the room code and building.	SR-4, SR-5	UI test
FR-9	The system shall provide a Room Details page showing scheduled classes for a selected room.	SR-4	UI/API test
FR-10	The Room Details page shall display class time, course name, and instructor when available.	SR-4	UI/API test
FR-11	The system shall provide an Admin Login page.	SR-7, SR-9	Authentication test
FR-12	The system shall allow an authenticated admin to upload a semester CSV file.	SR-7, SR-9	Upload test
FR-13	The backend shall reject CSV uploads with missing or invalid authorization tokens.	SR-9	Security test
FR-14	The backend shall parse uploaded CSV data and insert semester, room, and schedule records into MySQL.	SR-7	Database test
FR-15	The backend shall avoid creating duplicate room records during CSV upload.	SR-7	Database test
FR-16	The frontend shall display a disclaimer explaining that results are schedule-based suggestions.	SR-8	UI test
FR-17	The system shall support local and deployed environments using environment-based API URLs.	Project need	Deployment test
FR-18	The system shall exclude online and non-physical class locations from room search results.	SR-1, SR-8	Data test

6. Non-Functional Requirements

ID	Non-Functional Requirement	Verification
NFR-1	Typical room searches should return results within 3 seconds when the backend and database are online.	Performance check
NFR-2	The room search page should remain usable on desktop and smaller screens.	Responsive UI test

NFR-3	The frontend shall show a clear error message if a request fails.	Error handling test
NFR-4	The backend shall return JSON for room search and room schedule endpoints.	API test
NFR-5	Database credentials shall not be committed to GitHub.	Repository review
NFR-6	Firebase service account credentials shall not be committed to GitHub.	Repository review
NFR-7	Frontend and backend deployment values shall be stored using environment variables.	Configuration review
NFR-8	Admin upload requests shall require authentication before CSV data is processed.	Security test
NFR-9	Documentation shall explain how to run the frontend, backend, and database setup.	Documentation review
NFR-10	The system shall clearly communicate that results are not guaranteed real-time availability.	UI/documentation review

7. Architecture Requirements

ID	Architecture Requirement	Verification
AR-1	The system shall separate frontend, backend, and database responsibilities.	Architecture review
AR-2	The frontend shall be a React/Vite single-page app.	Code review
AR-3	The backend shall be a Flask API server.	Code review
AR-4	The database shall use MySQL.	Database review
AR-5	The frontend shall communicate with the backend through API endpoints only.	Code/API review
AR-6	The frontend shall not directly connect to MySQL.	Code review
AR-7	The deployed frontend shall use Vercel.	Deployment review
AR-8	The deployed backend shall use Render.	Deployment review
AR-9	The MySQL database shall be hosted on EC2.	Deployment review
AR-10	The backend shall configure CORS for approved frontend origins.	Security/deployment test

8. Security Requirements

ID	Security Requirement	Verification
SEC-1	Admin upload requests shall include an Authorization header with a Firebase bearer token.	API test
SEC-2	The backend shall verify the Firebase ID token before processing CSV uploads.	Backend test
SEC-3	Missing or invalid tokens shall return 401 Unauthorized.	Security test
SEC-4	Database credentials shall be stored in environment variables.	Configuration review
SEC-5	Firebase service account credentials shall not be committed to GitHub.	Repository review
SEC-6	Backend SQL queries shall use parameterized values for user-provided input.	Code review
SEC-7	CORS shall allow expected frontend origins to call the backend API.	Deployment test
SEC-8	Future versions should verify that an authenticated user is specifically an approved admin.	Future improvement

9. Use Cases

UC-1: Search for Suggested Rooms

Field	Description
Actor	Student/User
Goal	Find rooms that do not have scheduled class conflicts during a selected day and time range.
Main Flow	User selects day, time range, and building, submits search, and views results.
Success Result	The system displays schedule-based room suggestions.
Related Requirements	FR-1 through FR-8, FR-16, FR-18

UC-2: View Room Schedule Details

Field	Description
Actor	Student/User
Goal	View scheduled classes for a selected room.
Main Flow	User clicks View Room Schedule and views the selected room's schedule.
Success Result	The system displays class times, course name, and instructor.
Related Requirements	FR-9, FR-10

UC-3: Admin Login

Field	Description
Actor	Admin
Goal	Authenticate before accessing admin functions.
Main Flow	Admin enters email/password, Firebase authenticates the user, and the frontend stores the ID token.
Success Result	Admin can access protected admin workflow.
Related Requirements	FR-11, SEC-1, SEC-2, SEC-3

UC-4: Upload Semester Schedule Data

Field	Description
Actor	Admin
Goal	Upload semester CSV data so room searches use current schedule information.
Main Flow	Admin selects semester, chooses CSV, submits upload, backend verifies token, parses file, and stores data in MySQL.
Success Result	Semester, room, and schedule records are inserted into the database.
Related Requirements	FR-12 through FR-15

10. Assumptions and Constraints

ID	Assumption / Constraint
AC-1	The system uses class schedule data as the basis for room suggestions.
AC-2	A room with no scheduled class is not guaranteed to be physically available.
AC-3	The system does not currently track real-time room occupancy.
AC-4	The system depends on the accuracy of uploaded CSV schedule data.
AC-5	The current search design uses day and time, not full date-range scheduling.
AC-6	Admin upload requires Firebase Authentication to be correctly configured.
AC-7	The MySQL database must be reachable from the backend environment.
AC-8	Future features such as live occupancy, reservations, room pictures, notifications, and confidence scores are outside the current MVP.
